ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
Project Number _____

17.16IO Structural Checklist for Building Type PC2a: Precast Concrete Frames without Shear Walls

Table 17-33. Immediate Occupancy Structural Checklist for Building Type PC2a.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of moment frames in each principal direction is greater than or equal to two. The number of bays of moment frames in each line is greater than or equal to 3.	5.5.1.1	A.3.1.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN SHEAR STRESS CHECK: The shear stress in the concrete columns, calculated using the Quick Check procedure of Section 4.4.3.2 is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f_c'}$.	5.5.2.1.4	A.3.1.4.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN AXIAL STRESS CHECK: The axial stress caused by gravity loads in columns subjected to overturning forces is less than $0.10f_c'$. Alternatively, the axial stresses caused by overturning forces alone, calculated using the Quick Check procedure of Section 4.4.3.6 is less than $0.30f_c'$.	5.5.2.1.3	A.3.1.4.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PRECAST CONNECTION CHECK: The precast connections at frame joints have the capacity to resist the shear and moment demands calculated using the Quick Check procedure of Section 4.4.3.5.	5.5.2.4	A.3.1.5.1	
Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TOPPING SLAB: Precast concrete diaphragm elements are interconnected by a continuous reinforced concrete topping slab with a minimum thickness of 2 in. (50.8 mm).	5.6.4	A.4.5.1	

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Connections						
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TOPPING SLAB TO WALLS OR FRAMES: Reinforced concrete topping slabs that interconnect the precast concrete diaphragm elements are doweled for transfer of forces into the shear wall or frame elements, and the dowels are able to develop the least of the shear strength of the walls, frames, or slabs.	5.7.2	A.5.2.3
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	GIRDER–COLUMN CONNECTION: There is a positive connection using plates, connection hardware, or straps between the girder and the column support.	5.7.4.1	A.5.4.1

Status				**Evaluation Statement**	**Tier 2 Reference**	**Commentary Reference**	**Comments**
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PRESTRESSED FRAME ELEMENTS: The seismic-force-resisting frames do not include any prestressed or posttensioned elements where the average prestress exceeds the lesser of 700 lb/in.^{2} (4.83 MPa) or $f_c'/6$ at potential hinge locations. The average prestress is calculated in accordance with the Quick Check procedure of Section 4.4.3.8.	5.5.2.3.2	A.3.1.4.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CAPTIVE COLUMNS: There are no columns at a level with height-to-depth ratios less than 75% of the nominal height-to-depth ratio of the typical columns at that level.	5.5.2.3.3	A.3.1.4.5	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	JOINT REINFORCING: Beam–column joints have ties spaced at or less than $8d_b$.	5.5.2.3.8	A.3.1.4.13	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DEFLECTION COMPATIBILITY: Secondary components have the shear capacity to develop the flexural strength of the components.	5.5.2.5.2	A.3.1.6.2	
Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7	

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C	NC	N/A	U			
☐	☐	☐	☐	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8
Connections						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps; the pile cap reinforcement and pile anchorage are able to develop the tensile capacity of the piles.	5.7.3.5	A.5.3.8
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	GIRDERS: Girders supported by frames have at least two ties securing the anchor bolts unless provided with independent stiff wall anchors with strength to resist the connection force calculated in the Quick Check procedure of Section 4.4.3.7.	5.7.4.2	A.5.4.2
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	CORBEL BEARING: If the frame girders bear on column corbels, the length of bearing is greater than 3 in. (76 mm).	5.7.4.3	A.5.4.3
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	CORBEL CONNECTIONS: The frame girders are not connected to corbels with welded elements.	5.7.4.3	A.5.4.4
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	TRANSFER TO FRAMES: Diaphragms are connected for transfer of loads to the frames.	5.7.2	A.5.2.1

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate and High Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Foundation System							
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1	

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